007 (B01-B18) Mon

Q. No. Question Expected Time

to complete

1

Design and Development of a Simple Personal Portfolio

Decentralized Application (dApp) Frontend

Objective

To design and implement a simple decentralized application

(dApp) frontend for a Personal Portfolio Landing Page that

interacts with an Ethereum smart contract using HTML, CSS,

JavaScript, and Web3.js.

The dApp will:

• Connect to the user's MetaMask wallet

• Read portfolio information (name, role, and skills) stored

on the blockchain

• Display the data dynamically on a web page

Tools & Technologies Required

• VS Code (with Live Server extension recommended)

• Web browser (Chrome/Brave)

• MetaMask browser extension

• Node.js (optional, for local server)

• Ethereum test network (Sepolia / Ganache)

• Web3.js library

Prerequisites

• Basic knowledge of HTML, JavaScript

• Basic understanding of blockchain and smart contracts

• MetaMask wallet set up and connected to a test network

Code:

```
<!DOCTYPE html>

<html lang="en">
```

```
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Blockchain Engineering - Portfolio dApp</title>

    <!-- Tailwind CSS -->
    <script src="https://cdn.tailwindcss.com"></script>

    <!-- Chart.js -->
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

    <!-- Web3.js -->
    <script src="https://cdn.jsdelivr.net/npm/web3@4.5.0/dist/web3.min.js"></script>

    <!-- Chosen Palette: Slate & Emerald (Professional dark accents with Web3 green highlights) -->
    <!-- Application Structure Plan: Dual-purpose SPA.
        1. 'Assignment Brief' contains the exact constraints, tools, and the architectural model recreated with HTML/CSS.
        2. 'dApp Workspace' is the functional deliverable, ready to run in VS Code via Live Server. -->
    <!-- Visualization & Content Choices:
        - Architecture Model -> Goal: Understand -> Pure HTML/Tailwind CSS stacked layout to exactly replicate the provided image.
        - Developer Skills -> Goal: Visualize -> Radar Chart (Chart.js) fulfilling data viz requirements.
        - Web3 Interaction -> Goal: Execute -> Vanilla JS Web3 integration. -->
    <!-- CONFIRMATION: NO SVG graphics used. NO Mermaid JS used. -->

    <style>
        body {
            font-family: 'Segoe UI', Roboto, Helvetica, Arial, sans-serif;
            background-color: #f8fafc;
            color: #0f172a;
        }
```

```
.chart-container {
    position: relative;
    width: 100%;
    max-width: 500px;
    margin-left: auto;
    margin-right: auto;
    height: 300px;
    max-height: 350px;
    display: flex;
    justify-content: center;
    align-items: center;
}

@media (min-width: 768px) {
    .chart-container {
        height: 350px;
    }
}

.glass-panel {
    background: rgba(255, 255, 255, 0.95);
    backdrop-filter: blur(10px);
    border: 1px solid rgba(226, 232, 240, 0.8);
    box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.05);
}

/* Styles specifically to mimic the tags in the image */
.arch-tag {
    display: inline-block;
    background-color: #f1f5f9;
```

```
        color: #475569;
        font-family: monospace;
        font-size: 0.75rem;
        padding: 0.1rem 0.4rem;
        border-radius: 0.25rem;
        margin-top: 0.5rem;
      }

      ::-webkit-scrollbar { width: 6px; }
      ::-webkit-scrollbar-track { background: transparent; }
      ::-webkit-scrollbar-thumb { background: #cbd5e1; border-radius: 3px; }
      ::-webkit-scrollbar-thumb:hover { background: #94a3b8; }
  </style>
</head>
<body class="antialiased min-h-screen flex flex-col">

  <div id="toast-container" class="fixed bottom-5 right-5 z-50 flex flex-col gap-3"></div>

  <!-- Navigation -->
  <nav class="bg-slate-900 text-white sticky top-0 z-40 shadow-lg">
    <div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8">
      <div class="flex items-center justify-between h-16">
        <div class="flex items-center gap-3">
          <span class="text-2xl">&#x1F310;</span>
          <span class="font-bold text-lg tracking-wide">BE Practical-2</span>
        </div>
        <div class="flex space-x-4">
          <button onclick="switchTab('brief')" id="tab-brief" class="px-3 py-2 rounded-md text-sm font-medium bg-slate-800 text-white transition-colors hover:bg-slate-700">Assignment Brief</button>
```

```
            <button onclick="switchTab('workspace')" id="tab-workspace" class="px-3 py-2 rounded-
md text-sm font-medium text-slate-300 transition-colors hover:bg-slate-700 hover:text-white">dApp
Workspace</button>
          </div>
        </div>
      </div>
    </nav>

    <!-- Main Content -->
    <main class="flex-grow w-full max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-8">

      <!-- SECTION: Assignment Brief -->
      <section id="section-brief" class="block animate-fade-in">

        <h1 class="text-3xl font-extrabold text-slate-900 mb-6">Simple Personal Portfolio
Decentralized Application (dApp) Frontend</h1>

        <div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6 mb-8">
          <!-- Course Details -->
          <div class="glass-panel rounded-xl p-6 border-t-4 border-t-blue-500">
            <div class="flex items-center gap-2 mb-4">
              <span class="text-xl">&#x1F4DA;</span>
              <h2 class="text-xl font-bold text-slate-800">Course Details</h2>
            </div>
            <ul class="space-y-3 text-sm text-slate-700">
              <li class="flex justify-between border-b pb-1"><span class="font-
semibold">Course:</span> <span>Blockchain Engineering</span></li>
              <li class="flex justify-between border-b pb-1"><span class="font-
semibold">Code:</span> <span>23CS201PE401</span></li>
              <li class="flex justify-between border-b pb-1"><span class="font-
semibold">Year/Sem:</span> <span>III / II</span></li>
              <li class="flex justify-between border-b pb-1"><span class="font-
semibold">Instructor:</span> <span>Dr. Jagjit Singh Dhatterwal</span></li>
```

```
            <li class="flex justify-between pb-1"><span class="font-semibold">Assignment No:</span> <span>01/12</span></li>
          </ul>
        </div>

        <!-- Tools & Tech -->
        <div class="glass-panel rounded-xl p-6 border-t-4 border-t-emerald-500">
          <div class="flex items-center gap-2 mb-4">
            <span class="text-xl">&#x2692;&#xFE0F;</span>
            <h2 class="text-xl font-bold text-slate-800">Tools & Technologies Required</h2>
          </div>
          <ul class="list-disc list-inside space-y-2 text-sm text-slate-700">
            <li>VS Code (with Live Server extension recommended)</li>
            <li>Web browser (Chrome/Brave)</li>
            <li>MetaMask browser extension</li>
            <li>Node.js (optional, for local server)</li>
            <li>Ethereum test network (Sepolia / Ganache)</li>
            <li>Web3.js library</li>
          </ul>
        </div>

        <!-- Prerequisites -->
        <div class="glass-panel rounded-xl p-6 border-t-4 border-t-purple-500">
          <div class="flex items-center gap-2 mb-4">
            <span class="text-xl">&#x1F511;</span>
            <h2 class="text-xl font-bold text-slate-800">Prerequisites</h2>
          </div>
          <ul class="list-disc list-inside space-y-2 text-sm text-slate-700">
            <li>Basic knowledge of HTML, JavaScript</li>
            <li>Basic understanding of blockchain and smart contracts</li>
            <li>MetaMask wallet set up and connected to a test network</li>
```

```
            </ul>
          </div>
        </div>

        <!-- Objective Highlight -->
        <div class="bg-indigo-50 border border-indigo-100 rounded-xl p-6 shadow-sm mb-10">
          <h3 class="text-xl font-bold text-indigo-900 mb-3">&#x1F3AF; Objective</h3>
          <p class="text-indigo-800 leading-relaxed font-medium text-lg">
            To design and implement a simple decentralized application (dApp) frontend for a Personal Portfolio Landing Page that interacts with an Ethereum smart contract using HTML, CSS, JavaScript, and Web3.js.
          </p>
          <div class="mt-4 pt-4 border-t border-indigo-200">
            <h4 class="font-bold text-indigo-900 mb-2">The dApp will:</h4>
            <ul class="list-none text-indigo-800 space-y-1 ml-2 font-medium">
              <li>&bull; Connect to the user's MetaMask wallet</li>
              <li>&bull; Read portfolio information (name, role, and skills) stored on the blockchain</li>
              <li>&bull; Display the data dynamically on a web page</li>
            </ul>
          </div>
          <div class="mt-6">
            <button onclick="switchTab('workspace')" class="bg-indigo-600 hover:bg-indigo-700 text-white font-bold py-2 px-6 rounded-lg transition-colors shadow-md flex items-center gap-2">
              Launch dApp Workspace <span>&#x27A1;&#xFE0F;</span>
            </button>
          </div>
        </div>

        <!-- Architecture Model Recreation -->
        <div class="mb-10 w-full max-w-3xl mx-auto">
          <h3 class="text-2xl font-bold text-slate-800 mb-6 text-center border-b pb-2">dApp Architecture Model</h3>
```

```
<div class="flex flex-col items-start w-full relative">

  <!-- 1. Frontend -->
  <div class="w-full bg-white border border-slate-300 rounded-lg shadow-sm flex items-start p-4 relative border-l-[6px] border-l-blue-500 z-10">
    <div class="text-blue-500 text-3xl mr-4 mt-1">&#x1F4BB;</div> <!-- Monitor icon -->
    <div>
      <h4 class="font-bold text-slate-900 text-lg">Frontend User Interface</h4>
      <p class="text-sm text-slate-500 mt-1">User interaction point, displays data, initiates actions</p>
      <span class="arch-tag">React, Vue, IPFS Hosting</span>
    </div>
  </div>

  <!-- Arrow -->
  <div class="w-[60px] flex flex-col items-center">
    <div class="h-6 border-l-2 border-dashed border-slate-400"></div>
    <div class="text-slate-500 -mt-2.5 text-lg">&#x2193;</div>
  </div>

  <!-- 2. Wallet -->
  <div class="w-full bg-white border border-slate-300 rounded-lg shadow-sm flex items-start p-4 relative border-l-[6px] border-l-amber-500 z-10">
    <div class="text-amber-500 text-3xl mr-4 mt-1">&#x1F45B;</div> <!-- Wallet icon -->
    <div>
      <h4 class="font-bold text-slate-900 text-lg">Wallet / Signer</h4>
      <p class="text-sm text-slate-500 mt-1">Manages private keys, signs transactions, user authorization</p>
      <span class="arch-tag">MetaMask, Rainbow, EIP-1193</span>
    </div>
  </div>
```

```
<!-- Arrow -->
<div class="w-[60px] flex flex-col items-center">
  <div class="h-6 border-l-2 border-dashed border-slate-400"></div>
  <div class="text-slate-500 -mt-2.5 text-lg">&#x2193;</div>
</div>

<!-- 3. Provider -->
<div class="w-full bg-white border border-slate-300 rounded-lg shadow-sm flex items-start p-4 relative border-l-[6px] border-l-emerald-500 z-10">
  <div class="text-emerald-500 text-3xl mr-4 mt-1">&#x1F5A7;</div> <!-- Network icon -->
  <div>
    <h4 class="font-bold text-slate-900 text-lg">Provider (JSON-RPC)</h4>
    <p class="text-sm text-slate-500 mt-1">Gateway to blockchain network, standard API methods</p>
    <span class="arch-tag">Infura, Alchemy, eth_call</span>
  </div>
</div>

<!-- Arrow -->
<div class="w-[60px] flex flex-col items-center">
  <div class="h-6 border-l-2 border-dashed border-slate-400"></div>
  <div class="text-slate-500 -mt-2.5 text-lg">&#x2193;</div>
</div>

<!-- 4. Execution Layer -->
<div class="w-full bg-white border border-slate-300 rounded-lg shadow-sm flex items-start p-4 relative border-l-[6px] border-l-purple-500 z-10">
  <div class="text-purple-500 text-3xl mr-4 mt-1">&#x1F537;</div> <!-- Diamond/Ethereum-like icon -->
  <div>
    <h4 class="font-bold text-slate-900 text-lg">Execution Layer</h4>
```

```
                <p class="text-sm text-slate-500 mt-1">EVM Smart Contracts, deterministic logic execution</p>
                <span class="arch-tag">Solidity, Vyper, Bytecode</span>
              </div>
            </div>

            <!-- Arrow & Split lines -->
            <div class="w-[60px] flex flex-col items-center relative">
              <div class="h-6 border-l-2 border-dashed border-slate-400"></div>
            </div>

            <!-- 5. Storage & Indexing (Split) -->
            <div class="w-full grid grid-cols-1 sm:grid-cols-2 gap-4">

              <!-- Storage -->
              <div class="bg-white border border-slate-300 rounded-lg shadow-sm flex items-start p-4 border-l-[6px] border-l-red-500">
                <div class="text-red-500 text-3xl mr-4 mt-1">&#x1F6E2;&#xFE0F;</div> <!-- Database/Drum icon -->
                <div>
                  <h4 class="font-bold text-slate-900 text-lg">Storage</h4>
                  <p class="text-sm text-slate-500 mt-1">State & Assets</p>
                  <span class="arch-tag">On-chain, IPFS</span>
                </div>
              </div>

              <!-- Indexing -->
              <div class="bg-white border border-slate-300 rounded-lg shadow-sm flex items-start p-4 border-l-[6px] border-l-pink-500">
                <div class="text-pink-500 text-3xl mr-4 mt-1">&#x1F50D;</div> <!-- Magnifying glass icon -->
                <div>
                  <h4 class="font-bold text-slate-900 text-lg">Indexing</h4>
```

```
                <p class="text-sm text-slate-500 mt-1">Query Layer</p>
                <span class="arch-tag">The Graph</span>
              </div>
            </div>

          </div>
        </div>
      </div>
    </section>

    <!-- SECTION: dApp Workspace -->
    <section id="section-workspace" class="hidden animate-fade-in">
      <div class="mb-6 flex flex-col md:flex-row md:items-center justify-between gap-4">
        <div>
          <h1 class="text-3xl font-extrabold text-slate-900 mb-2">Personal Portfolio dApp</h1>
          <p class="text-slate-600">Frontend interface connected to Web3.js</p>
        </div>

        <div class="flex flex-col items-end gap-2 bg-white p-3 rounded-lg border shadow-sm">
          <div class="flex items-center gap-3">
            <div id="connection-status-dot" class="w-3 h-3 rounded-full bg-red-500"></div>
            <span id="connection-status-text" class="text-sm font-semibold text-slate-700">Not Connected</span>
            <button id="btn-connect" class="bg-emerald-600 hover:bg-emerald-700 text-white text-sm font-bold py-2 px-4 rounded transition-colors flex items-center gap-2 shadow-sm">
              <span class="text-lg">&#x1F98A;</span> Connect Wallet
            </button>
          </div>
          <div id="wallet-address-display" class="text-xs text-slate-500 font-mono bg-slate-100 px-2 py-1 rounded hidden border"></div>
        </div>
      </div>
```

```
<!-- Portfolio UI -->
<div id="portfolio-ui" class="bg-white rounded-2xl shadow-xl border border-slate-200 overflow-hidden opacity-50 transition-opacity duration-500 relative">

  <!-- Overlay -->
  <div id="connect-overlay" class="absolute inset-0 bg-slate-900/10 backdrop-blur-sm flex flex-col items-center justify-center z-10 transition-opacity duration-300">
    <div class="bg-white p-6 rounded-xl shadow-lg text-center max-w-sm border-t-4 border-emerald-500">
      <span class="text-4xl block mb-3">&#x1F512;</span>
      <h3 class="font-bold text-lg text-slate-800 mb-2">Web3 Connection Required</h3>
      <p class="text-sm text-slate-600 mb-4">Connect MetaMask to read portfolio data from the Ethereum blockchain.</p>
      <button onclick="toggleMockMode()" class="text-xs text-indigo-600 hover:underline">
        Or load mock data for frontend testing
      </button>
    </div>
  </div>

  <div class="grid grid-cols-1 lg:grid-cols-12">
    <!-- Profile Info -->
    <div class="lg:col-span-5 bg-gradient-to-br from-slate-900 to-slate-800 p-8 text-white flex flex-col justify-center items-center text-center">
      <div class="w-32 h-32 bg-slate-700 rounded-full border-4 border-emerald-400 mb-6 flex items-center justify-center shadow-lg">
        <span class="text-6xl">&#x1F468;&#x200D;&#x1F4BB;</span>
      </div>
      <h2 id="display-name" class="text-3xl font-bold mb-2 tracking-tight">--------</h2>
      <div class="bg-emerald-500/20 text-emerald-300 px-4 py-1.5 rounded-full text-sm font-semibold tracking-wider uppercase mb-6 border border-emerald-500/30">
        <span id="display-role">--------</span>
      </div>
```

```
            <p class="text-slate-400 text-sm leading-relaxed max-w-xs">
              Decentralized professional portfolio verified on-chain.
            </p>
          </div>

          <!-- Skills Viz -->
          <div class="lg:col-span-7 p-8">
            <h3 class="text-xl font-bold text-slate-800 border-b pb-3 mb-6 flex items-center gap-2">
              <span>&#x2699;&#xFE0F;</span> Verified Skills
            </h3>

            <div id="skills-list" class="flex flex-wrap gap-2 mb-8 min-h-[40px]">
              <span class="text-slate-400 italic text-sm">Waiting for blockchain data...</span>
            </div>

            <div class="bg-slate-50 rounded-xl p-4 border border-slate-100">
              <h4 class="text-center text-sm font-semibold text-slate-500 mb-2 uppercase tracking-
wider">Skill Proficiency Analysis</h4>
              <div class="chart-container">
                <canvas id="skillsChart"></canvas>
              </div>
            </div>
          </div>
        </div>
      </div>

      <div class="mt-8 bg-amber-50 border border-amber-200 rounded-lg p-5">
        <h4 class="font-bold text-amber-900 mb-2 flex items-center gap-2">
          <span>&#x1F6E0;&#xFE0F;</span> VS Code Execution Instructions
        </h4>
        <ol class="list-decimal list-inside text-sm text-amber-900 space-y-2 font-mono bg-white p-3
rounded border border-amber-100">
```

```
            <li>Ensure you have the <span class="font-bold">Live Server</span> extension installed in
VS Code.</li>

            <li>Right-click this HTML file and select <span class="font-bold">"Open with Live
Server"</span>.</li>

            <li>If you have a deployed contract, update <code class="bg-amber-100 px-1
rounded">CONTRACT_ADDRESS</code> and <code class="bg-amber-100 px-1
rounded">CONTRACT_ABI</code> in the script below.</li>

            <li>If testing without a contract, click "Or load mock data for frontend testing" inside the
dApp Workspace.</li>

          </ol>

        </div>

      </section>


    </main>


    <script>

      // --- UI & NAVIGATION ---

      function switchTab(tabName) {

        document.getElementById('tab-brief').className = tabName === 'brief'

          ? 'px-3 py-2 rounded-md text-sm font-medium bg-slate-800 text-white transition-colors
hover:bg-slate-700'

          : 'px-3 py-2 rounded-md text-sm font-medium text-slate-300 transition-colors hover:bg-
slate-700 hover:text-white';


        document.getElementById('tab-workspace').className = tabName === 'workspace'

          ? 'px-3 py-2 rounded-md text-sm font-medium bg-slate-800 text-white transition-colors
hover:bg-slate-700'

          : 'px-3 py-2 rounded-md text-sm font-medium text-slate-300 transition-colors hover:bg-
slate-700 hover:text-white';


        document.getElementById('section-brief').classList.toggle('hidden', tabName !== 'brief');

        document.getElementById('section-workspace').classList.toggle('hidden', tabName !==
'workspace');
```

```
    if (tabName === 'workspace' && skillsChartInstance) {
        skillsChartInstance.resize();
    }
}

function showMessage(title, message, type = 'info') {
    const container = document.getElementById('toast-container');
    const toast = document.createElement('div');
    let borderClass = type === 'error' ? 'border-red-500' : type === 'success' ? 'border-emerald-500' : 'border-blue-500';
    let icon = type === 'error' ? '&#x26A0;&#xFE0F;' : type === 'success' ? '&#x2705;' : '&#x2139;&#xFE0F;';

    toast.className = `bg-white border-l-4 ${borderClass} shadow-lg rounded p-4 min-w-[300px] animate-[slideIn_0.3s_ease-out]`;
    toast.innerHTML = `
        <div class="flex items-start gap-3">
            <span class="text-xl">${icon}</span>
            <div>
                <h4 class="font-bold text-slate-800 text-sm">${title}</h4>
                <p class="text-slate-600 text-xs mt-1">${message}</p>
            </div>
        </div>`;
    container.appendChild(toast);
    setTimeout(() => { toast.style.opacity = '0'; toast.style.transition = 'opacity 0.3s ease'; setTimeout(() => toast.remove(), 300); }, 4000);
}

// --- CHART.JS ---
let skillsChartInstance = null;

function renderChart(skillsData) {
```

```
const ctx = document.getElementById('skillsChart').getContext('2d');
if (skillsChartInstance) skillsChartInstance.destroy();

const labels = skillsData.map(skill => skill.length > 16 ? skill.substring(0, 13) + '...' : skill);
const dataScores = skillsData.map(() => Math.floor(Math.random() * (95 - 75 + 1)) + 75);

skillsChartInstance = new Chart(ctx, {
    type: 'radar',
    data: {
        labels: labels,
        datasets: [{
            label: 'Skill Proficiency (%)',
            data: dataScores,
            backgroundColor: 'rgba(16, 185, 129, 0.2)',
            borderColor: 'rgba(16, 185, 129, 1)',
            pointBackgroundColor: 'rgba(15, 23, 42, 1)',
            pointBorderColor: '#fff',
            pointHoverBackgroundColor: '#fff',
            pointHoverBorderColor: 'rgba(16, 185, 129, 1)',
            borderWidth: 2
        }]
    },
    options: {
        responsive: true,
        maintainAspectRatio: false,
        scales: {
            r: {
                angleLines: { color: 'rgba(0,0,0,0.1)' },
                grid: { color: 'rgba(0,0,0,0.1)' },
                pointLabels: { font: { family: "'Segoe UI', sans-serif", size: 12 }, color: '#475569' },
                ticks: { display: false, min: 0, max: 100 }
```

```
                }
            },
            plugins: {
                legend: { display: false },
                tooltip: {
                    backgroundColor: 'rgba(15, 23, 42, 0.9)',
                    padding: 10,
                    callbacks: { label: context => ` Proficiency: ${context.parsed.r}%` }
                }
            }
        }
    });
}

// --- WEB3.JS ---
const CONTRACT_ADDRESS = "0x0000000000000000000000000000000000000000"; // STUDENT: Update this!
const CONTRACT_ABI = [
    {
        "inputs": [],
        "name": "getPortfolio",
        "outputs": [
            { "internalType": "string", "name": "name", "type": "string" },
            { "internalType": "string", "name": "role", "type": "string" },
            { "internalType": "string[]", "name": "skills", "type": "string[]" }
        ],
        "stateMutability": "view",
        "type": "function"
    }
];
```

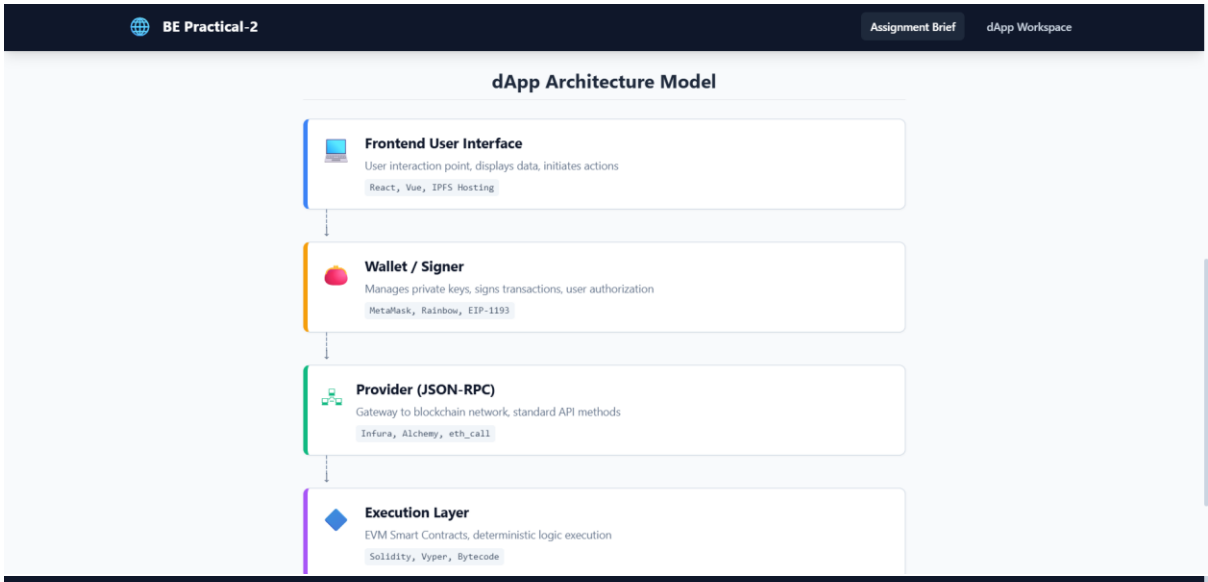
```
let web3;
let userAccount = null;
let isMockMode = false;

async function connectWallet() {
  if (isMockMode) return showMessage("Info", "Running in mock data mode.", "info");

  if (typeof window.ethereum !== 'undefined') {
    try {
      const accounts = await window.ethereum.request({ method: 'eth_requestAccounts' });
      userAccount = accounts[0];
      web3 = new Web3(window.ethereum);

      updateUIConnected();
      showMessage("Success", "MetaMask Wallet Connected!", "success");
      await fetchPortfolioData();
    } catch (error) {
      showMessage("Connection Error", "User denied account access.", "error");
    }
  } else {
    showMessage("MetaMask Missing", "Please install MetaMask to use this dApp.", "error");
  }
}

function updateUIConnected() {
  document.getElementById('connection-status-dot').classList.replace('bg-red-500', 'bg-emerald-500');
  document.getElementById('connection-status-text').innerText = "Connected";
  document.getElementById('connection-status-text').classList.replace('text-slate-700', 'text-emerald-700');

  const addrDisplay = document.getElementById('wallet-address-display');
```

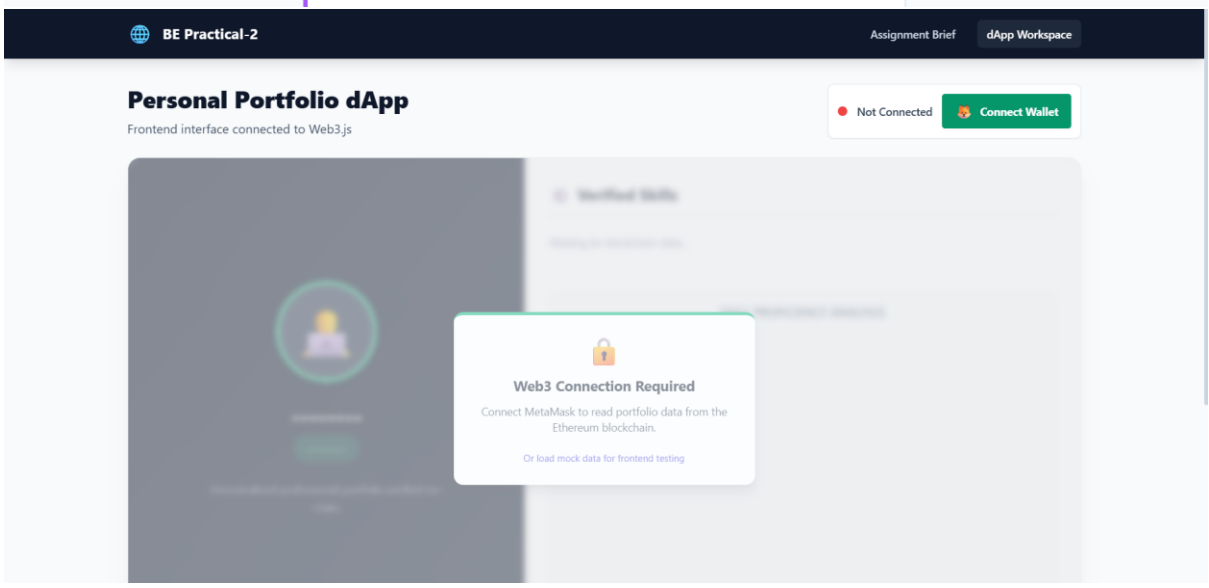
```
        addrDisplay.innerText =
`${userAccount.substring(0,6)}...${userAccount.substring(userAccount.length - 4)}`;
        addrDisplay.classList.remove('hidden');

        document.getElementById('btn-connect').innerText = "Refresh Data";
        document.getElementById('connect-overlay').style.opacity = '0';
        setTimeout(() => {
            document.getElementById('connect-overlay').classList.add('hidden');
            document.getElementById('portfolio-ui').classList.replace('opacity-50', 'opacity-100');
        }, 300);
    }

    async function fetchPortfolioData() {
        try {
            const contract = new web3.eth.Contract(CONTRACT_ABI, CONTRACT_ADDRESS);
            const data = await contract.methods.getPortfolio().call();
            populatePortfolio(data.name, data.role, data.skills);
        } catch (error) {
            showMessage("Contract Error", "Failed to fetch from contract. Loading mock data.",
"error");
            toggleMockMode(true);
        }
    }

    function populatePortfolio(name, role, skillsArray) {
        document.getElementById('display-name').innerText = name || "Unnamed User";
        document.getElementById('display-role').innerText = role || "Role Undefined";

        const skillsContainer = document.getElementById('skills-list');
        skillsContainer.innerHTML = '';

        if(skillsArray && skillsArray.length > 0) {
```

```
            skillsArray.forEach(skill => {
                const badge = document.createElement('span');
                badge.className = 'bg-slate-800 text-slate-100 px-3 py-1.5 rounded-md text-sm font-
medium border border-slate-700 shadow-sm';
                badge.innerText = skill;
                skillsContainer.appendChild(badge);
            });
            renderChart(skillsArray);
        } else {
            skillsContainer.innerHTML = '<span class="text-slate-400 italic text-sm">No skills
found.</span>';
        }
    }

    function toggleMockMode(forced = false) {
        isMockMode = true;
        userAccount = "0x89205A3A3b2A69De6Dbf7f01ED13B2108B2c43e7";
        updateUIConnected();
        populatePortfolio("Alice Blockchain", "Smart Contract Developer", ["Solidity", "Web3.js",
"React", "Hardhat", "Ether.js"]);
        if(!forced) showMessage("Mock Mode", "Loaded testing data.", "info");
    }

    document.getElementById('btn-connect').addEventListener('click', connectWallet);

    if (window.ethereum) {
        window.ethereum.on('accountsChanged', (accounts) => {
            if (accounts.length === 0) location.reload();
            else {
                userAccount = accounts[0];
                updateUIConnected();
                if(!isMockMode) fetchPortfolioData();
```

```
            }
        });
    }

    // Tailwind animations setup
    tailwind.config = {
        theme: {
            extend: {
                keyframes: {
                    'fade-in': { '0%': { opacity: '0', transform: 'translateY(10px)' }, '100%': { opacity: '1',
transform: 'translateY(0)' } },
                    slideIn: { '0%': { transform: 'translateX(100%)', opacity: '0' }, '100%': { transform:
'translateX(0)', opacity: '1' } }
                },
                animation: { 'fade-in': 'fade-in 0.4s ease-out forwards' }
            }
        }
    }
  </script>
</body>
</html>
```

Output:


BE Practical-2
Assignment Brief
dApp Workspace
dApp Architecture Model
Frontend User Interface
User interaction point, displays data, initiates actions
React, Vue, IPFS Hosting
Wallet / Signer
Manages private keys, signs transactions, user authorization
MetaMask, Rainbow, EIP-1193
Provider (JSON-RPC)
Gateway to blockchain network, standard API methods
Infura, Alchemy, eth_call
Execution Layer
EVM Smart Contracts, deterministic logic execution
Solidity, Vyper, Bytecode


BE Practical-2
Assignment Brief
dApp Workspace
Personal Portfolio dApp
Frontend interface connected to Web3.js
Not Connected
Connect Wallet
Web3 Connection Required
Connect MetaMask to read portfolio data from the Ethereum blockchain.
Or load mock data for frontend testing